

Article

Plasma Gasification of Medical Plastic Waste to Syngas in a Greenhouse Gas (CO₂) Environment

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Abstract: The global coronavirus (COVID-19) pandemic in early 2020 caused the amount of medical waste, especially plastic waste, to increase. The pandemic exacerbated the plastic waste management problem, including the need to find more sustainable treatment methods. This study investigated the sustainable conversion of plastic waste (FFP2-type face masks) to syngas via pure CO₂ plasma gasification to recover energy and reduce environmental pollution. A direct current (DC) thermal arc plasma torch of 40.6–68.4 kW power generated the plasma stream. Carbon dioxide (CO₂), as a greenhouse gas (GHG), was used as the main plasma-forming gas and gasifying agent. The 140_{thermal} feedstock input plasma gasification system was used in the study. The effect of the CO₂-to-C ratio on the gasification performance efficiency was investigated. The best CO₂ plasma gasification process performance was obtained at a CO₂-to-C ratio of 2.34. In these conditions, the main syngas components (H₂ + CO) comprised 80.46 vol.% (H₂: 24.62 vol.% and CO: 55.84 vol.%) and the following values were seen for the heating value of the syngas (LHV_{syngas}: 13.88 MJ/Nm³), the syngas yield (3.13 Nm³/kg_{FFP2}), the tar content in the syngas (23.0 g/Nm³), the carbon conversion efficiency (CCE: 70.6%), and the cold gas efficiency (CGE: 47.8%). Additionally, the plasma gasification process mass and energy balance were evaluated. It was demonstrated that CO₂ plasma gasification could be a promising thermochemical treatment technology for sustainable plastic waste disposal and the simultaneous utilization of greenhouse gases, such as carbon dioxide.

Keywords: sustainable waste management; FFP2 medical face mask; thermal arc plasma; greenhouse gas (CO₂); syngas; mass and energy balance



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1. Introduction

The global coronavirus (COVID-19) pandemic in early 2020 caused the amount of medical waste to increase. Overall, the pandemic has increased medical waste management problems, including the need for sustainable treatment options. According to the official statistics in the United Nations Environmental Programme report, the medical waste generated by healthcare facilities as a result of COVID-19 has increased to 3.4 kg/person/day compared to the ‘normal’ average volume of medical hazardous waste, which ranges from 0.2 to 0.5 kg/person/day [1]. The main types of COVID-19-related medical waste include masks, gloves, gowns, antigen tests, vaccine vials and needles, and plastic packaging

and containers, with most of this waste comprising polymers (plastics, rubbers, clothing, etc.) [2,3]. Also, the public use of personal protective equipment, especially masks, has increased significantly since the pandemic. Some estimations show that in 2020, up to 3.4 billion single-use masks were discarded daily, resulting in a considerable increase in the volume of plastic waste [4,5]. However, after the suppression of the first global wave of COVID-19, the use of protective masks in public (but not in the healthcare sector) has sharply decreased. Nevertheless, problems surrounding sustainable waste management, especially of plastic waste, and environmental pollution still exist. The World Health Organization (WHO), in its 2020 report, states that three out of ten healthcare facilities lack medical waste segregation systems. Also, fewer than one in three healthcare facilities in the least developed countries have a basic waste management service [6,7]. The healthcare sector makes a substantial contribution to greenhouse gas emissions, accounting for approx. 4–5% of total emissions [8,9]. Thus, COVID-19-related issues, including post-COVID-19, have made medical waste generation a global challenge [10].

Current medical waste management practice is based on the 3Rs (reduce, reuse, and recycle) concept. However, further treatment and disposal of medical waste is unavoidable to manage it safely and rationally. The three main options for treating medical waste are chemical treatment, autoclaving, and incineration, which are further divided into subcategories [11,12]. The residual waste remaining after any of the treatment options mentioned above is mainly disposed of in landfills. In this regard, more advanced medical waste treatment methods have been investigated over the years. Besides the use of medical waste incineration to produce power and heat, conventional thermochemical methods, such as pyrolysis and gasification, enable the conversion of waste to higher-added-value products, for instance, syngas, bio-oil, and biochar. Catalysts are usually utilized to overcome any issues with the final product quality improvement, energy consumption, and process cost reduction [13,14]. However, this causes additional problems, requiring catalyst recovery due to the deactivation of the catalyst by unavoidable side product streams such as tars, the need for expensive additional gas cleaning systems, or special waste pre-treatment requirements.

Plasma gasification for waste treatment has been considered as an advanced and more sustainable treatment method that enables, due to the very high temperatures (2000–5000 °C), the conversion of (hazardous) waste to gaseous compounds and vitrified solid inert slag [15–17]. Plasma offers a unique charged particle medium, such as electrons, ions, radicals, etc., which accelerates the conversion reaction kinetics and chemistry much faster than conventional gasification methods. It has been applied to several waste feedstock conversions, such as biomass, municipal solid waste (MSW), refuse-derived fuels (RDF), sewage sludge, medical waste, and plastic waste [17–23]. Various gases and their mixtures, for instance, air, nitrogen, argon, steam, carbon dioxide, etc., can be used to form plasma. Corresponding to global warming, direct carbon dioxide utilization via the carbon capture and utilization (CCU) process could be one of the solutions, as carbon dioxide can be used simultaneously as the gasifying agent, the main plasma gas, and the heat carrier, thus contributing to lowering emissions [24–27].

In this regard, to contribute to solving waste management problems, this experimental study aimed to investigate the possibility of gasifying medical plastic waste, i.e., FFP2-type face masks, to syngas by DC thermal arc plasma technology using carbon dioxide as the plasma-forming gas, the heat carrier, and the reactant. The performance of the plasma gasification system was quantified based on key process parameters, such as the producer gas composition and quality, the carbon conversion efficiency, the cold gas efficiency, and the specific energy requirement. Additionally, the process mass and energy balance were evaluated.

2. Materials and Methods

2.1. Plastic Feedstock Characterization, Pre-Treatment, and Dosing

The FFP2 face mask comprises four layers of fabric, with polypropylene (PP) and polyethylene (PE) used as the basic materials for the filter layers. Table 1 describes the proximate and ultimate analysis of the FFP2 face mask material used as a feedstock for the thermal plasma gasification to syngas.

Table 1. Proximate and ultimate analysis of the FFP2 pellets.

Proximate Analysis (wt.%)		Ultimate Analysis (wt.%)	
Moisture	0.15 ± 0.01	C	85.19 ± 0.68
Volatile matter	96.70 ± 1.05	H	14.58 ± 0.07
Fixed carbon	n.d.	N	<0.01
Ash	3.06 ± 0.09	S	0.027 ± 0.008
HHV, MJ/kg	51.54 ± 0.16	O	n.d.
LHV, MJ/kg	48.44 ± 0.16	Cl	0.008 ± 0.004

The FFP2 fabric was shredded to pieces and later granulated to pellets measuring 8.0 ± 0.5 mm in diameter (one pellet weighed around 0.5 ± 0.005 g). This pre-treatment step ensured a stabilized dosage of the feedstock from a 70 l hopper via a screw feeder to the reactor, regulated by a frequency converter. The reactor's average constant feeding rate of FFP2 granules was around 10.58 kg/h. Before the experiments, the pellet dosing was calibrated by checking the average mass flow rate, as the screw feeder, in general, cannot provide very smooth continuous dosing. Additionally, the mean mass flow rate of the FFP2 pellets fed to the plasma gasifier per measured unit time was checked after the experiments by weighing the remaining feedstock in the hopper. The difference from the initial tests was below 8%.

2.2. Plasma Gasifier

The experimental plasma gasification system, shown in Figure 1, was designed and built at the Plasma Processing Laboratory of the Lithuanian Energy Institute. The 140 kW_{th} input plasma gasifier with a reactor length of 1.3 m and an inner diameter of 0.2 m (reactor volume 0.04 m³) was used to carry out the experiments. The inner reactor walls were insulated with a 25 mm thick ceramic material made of aluminum oxide.

The main parts of the system consist of the following: 1—an atmospheric pressure DC arc plasma torch, 2—a feedstock hopper with a screw feeder, 3—a cyclone, 4—a gas quencher, 5—a gas burner, 6—a rotameter, 7—a gas and tar sampling port, 8—a condensate removal system, 9—a plasma–chemical reactor, 10—an ash/char hopper, 11—a power supply, 12—a plasma-forming and -shielding gas supply, 13—a plasma torch cooling system, and T1, T2, T3, T4, T5, T6, T7—thermocouples.

The DC thermal arc plasma torch, installed at the bottom of the reactor, of 40.6–68.4 kW power ($I = 160\text{--}190$ A, $U = 250\text{--}380$ V, $\eta = 0.58\text{--}0.73$) operating at atmospheric pressure served as a device to heat the carbon dioxide (CO₂) to an average temperature of $3250\text{--}3650 \pm 50$ K. Carbon dioxide, known as a GHG, was simultaneously used as the plasma-forming gas, the heat carrier, and the reactant. No other gas was used to form the plasma. The mass flow rate of the CO₂ was in the range of 11.56–21.13 kg/h. The FFP2 pellets dropped from the top of the reactor came into direct contact with a high-temperature CO₂ plasma stream on a moving grate, where it was converted to gaseous products. Below the moving grate was a bunker for the unconverted solid char/ash collection.

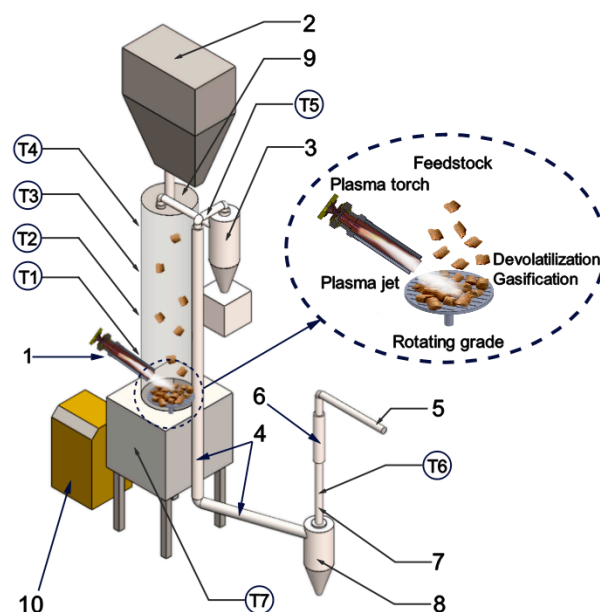


Figure 1. Plasma gasification system.

In situ gas product analysis was performed with a SWG 300^{−1} gas analyzer (MRU Instruments, USA) and subsequent sampling to the gas backs, which was later transported to an Agilent 7890A gas chromatograph (Agilent Technologies, Germany) with a dual-channel thermal conductivity detector and a valve system. The front channel with a pure He-carrying gas (10 mL/min) was used for separating the O₂, CO, CO₂, CH₄, and C₂H₂, whereas the H₂ was separated through a molecular-sieve packed column of the back channel operating on an N₂-carrier (30 mL/min). Three samples were taken for each experimental point at steady-state conditions. Each experimental point was measured for around 20 min.

The tar concentration measurement in the producer gas was determined using a standard cold-trapping method [28]. Tar condensation was performed in a solvent isopropanol (99.5%). After the condensation, the obtained tar compounds were analyzed with a Varian GC-3800 gas chromatograph (PerkinElmer, USA) equipped with a flame ionization detector (FID). The compounds were identified by examining the calibration mixture (EPA 610). More details can be found in [29].

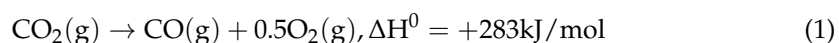
2.3. Plasma Gasification Performance Evaluation

To determine the plasma gasifier's performance, it was quantified in terms of the producer gas composition, the H₂ and CO yield, the H₂/CO ratio, the syngas lower heating value (LHV_{syngas}), the tar content in the syngas, the carbon conversion efficiency (CCE), the cold gas efficiency (CGE), and the specific energy requirement (SER), according to the methodology described in [30,31].

3. Results and Discussion

3.1. Effect of CO₂ as a Plasma-Forming Gas and Gasifying Agent

Firstly, the experiments were started by igniting the plasma torch, operating on CO₂ without FFP2 feedstock, to preheat the reactor. During the preheating phase, pure CO₂ was used as the plasma-forming gas and later as a gasifying agent. With the assistance of an electric arc, it was split into carbon monoxide and oxygen according to reaction (R1) [32].



Thermal CO₂ splitting is a highly endothermic reaction. However, at temperatures between 3000 K and 3500 K, thermal CO₂ splitting is thermodynamically and energetically favorable because of the higher-equilibrium CO₂ conversion of 45–80% and the lower reaction enthalpy of 206 kJ/mol [32]. In this temperature range, the theoretical energy efficiency is around 47%, which goes down to 35% at 5000 K, even if the CO₂ conversion reaches 100% [33].

Figure 2 shows the change in the CO₂ concentration over time, measured during the reactor preheating phase and later during the gasification. The average CO₂ plasma stream temperature calculated by the calorimetric method at the plasma torch's exit nozzle was up to 6000 ± 50 K, and the plasma torch power was 40.6–68.4 kW.

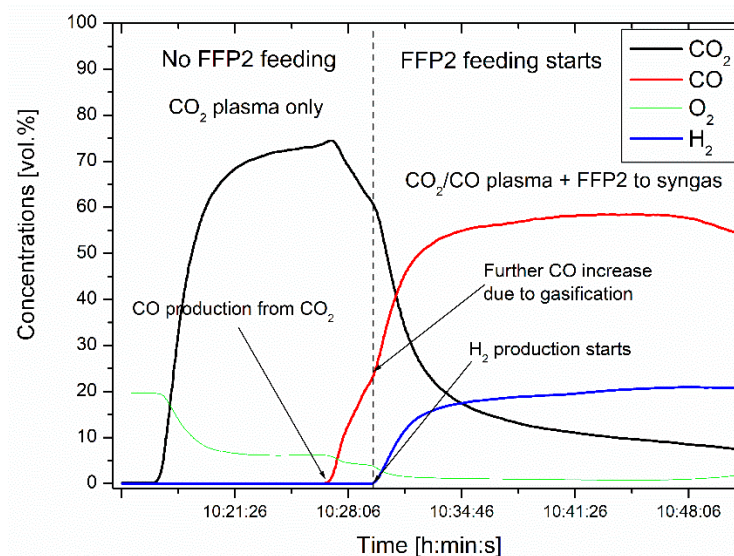


Figure 2. Gas concentration changes during the plasma gasifier preheating step with no feeding of the FFP2 pellets, and later during gasification with feeding of the FFP2 pellets.

It can be seen that during the gasifier preheating step (until 10:29:30), the CO₂ concentration decreased from 74% to around 60% and the CO concentration increased to 23% due to the CO₂-splitting reaction (R1). This means that one of the syngas constituents, carbon monoxide, was formed during the thermal CO₂ decomposition by the electric arc of the plasma torch. Due to some technical limitations in heating the reactor and avoiding the melting of the grate, the reactor was not in an entirely steady state, reaching the maximum CO concentrations after the CO₂ splitting. However, it was close to a steady-state condition, as observed during several experiments. After making some reactor modifications, this phenomenon will be thoroughly examined to investigate the highest possible CO concentration and compare it with the theoretical CO₂ conversion and energy efficiency. After the gasifier's preheating step with CO production via CO₂ splitting, the plasma gasification of the FFP2 pellets took over. In this case, gasification reactions occurred, enhancing the CO production to a total volumetric concentration of around 53–56% and further reducing the CO₂ to 6–8%. Thus, the CO production was affected by two dominant processes, i.e., the CO₂ splitting and the major chemical gasification reactions between the CO₂/CO plasma stream and the FFP2 feedstock (thermal cracking, Boudouard, partial oxidation, and other chemical reactions) [26,32,34].

3.2. Effect of the CO₂-to-Carbon (CO₂-to-C) Ratio on the Gasification Process Performance

During the plasma gasification step of the conversion of the FFP2 pellets to syngas, the effect of the CO₂ to carbon present in the feedstock (CO₂-to-C ratio) was investigated. As the mass flow of the FFP2 was kept constant at 10.58 kg/h, the carbon content did not

change over time. The main varying parameter was the mass flow of the CO₂ (the only gasifying medium and plasma-forming gas), which was in the 11.56–21.13 kg/h range. It should be noted that changing the mass flow of the CO₂ also affected the power of the plasma torch due to a voltage drop in the electric arc discharge chamber. Thus, by increasing the CO₂ flow rate from 11.56 to 21.13 kg/h, the plasma torch power increased from 40.6 to 68.4 kW and more energy was transferred to the reactor.

As the CO₂-to-C ratio increased from 1.26 to 2.34, the volumetric concentrations of the gaseous components in the producer gas changed, as shown in Figure 3. The major gas component was CO (52.86–55.84 vol.%), followed by H₂ (20.61–24.62 vol.%), CO₂ (6.55–8.24 vol.%), CH₄ (5.83–7.64 vol.%), C₂H₂ (~3 vol.%), and traces of other light C₂–C₃ hydrocarbons. The reason for the increasing CO concentration may be the dominance of the CO₂-splitting (R1) and Boudouard reactions (R2), while the production of H₂ can be explained by the feedstock thermal cracking (R3) and the water-gas shift reactions (R4). Thermal plasma provides a high-temperature region for partial oxidation, cracking, reforming, and other reactions when directly contacting the plastic pellets at short residence times. Therefore, the cracking process at high temperatures resulted in higher volumes of gaseous products, such as H₂, and less coke formation [35,36]. There were also several reactions involving methane formation: (R5), (R6), and (R7). Syngas (H₂ + CO) generally constituted more than 76.08–80.46 vol.% in the producer gas.

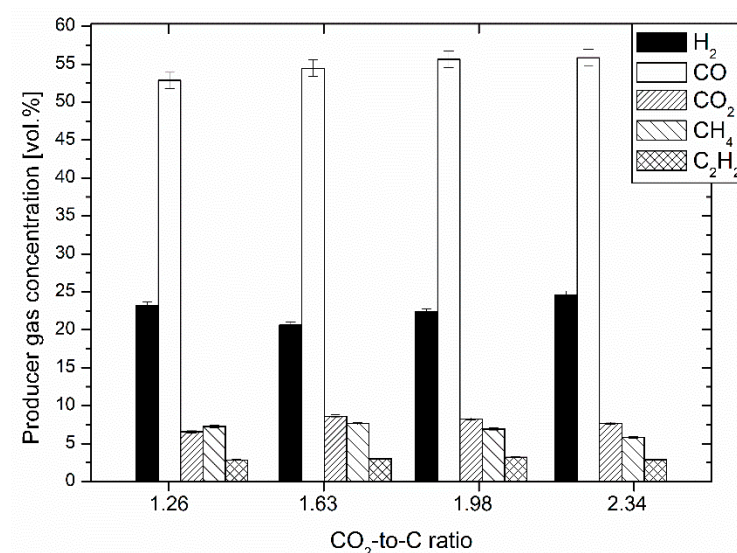
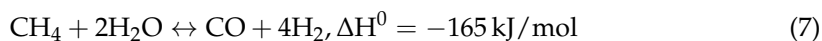
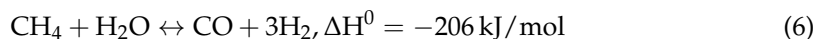
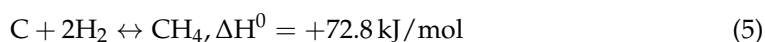
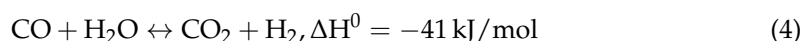
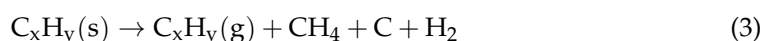
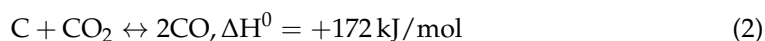


Figure 3. Effect of the CO₂-to-C ratio on producer gas content.

The concentration of CO (~53–56%) in this study was higher than that reported by other authors: Agon et al. [18] (CO: 42.5–46%) and Mallick and Vairakannu [25] (CO: 41.59–44.06%)—gasifying RDF, Mallick and Vairakannu [26]—gasifying electrical switch waste comprised of bakelite (CO: 40.27–43.61%), and Mallick and Vairakannu [37]—

gasifying computer keyboard plastic waste (CO: 49.3–51.98%) using CO₂ as a gasifying medium. The higher CO concentration in this study could be due to the pure CO₂ used as the primary plasma gas, which had a longer residence time when passing and contacting the electric arc of the plasma torch and thus splitting into CO and O₂. Besides, instead of pure CO₂, a mixture of CO₂ + O₂ + Ar as the gasifying medium and plasma gas was reported in [18]. Mallick and Vairakannu and Li et al. [25,26,37] reported using pure CO₂ plasma gasification of the selected feedstock in the 1.12–2.0 kW plasma torch power range. The CO concentration was around 52% in the computer keyboard plastic waste conversion case, but not at the optimal process conditions. Sikarwar et al. [38] also reported a similar CO concentration of around 57 vol.%. However, CO₂ was used only as the gasifying agent, whereas Ar + H₂O was used as the plasma-forming gas, yielding a higher H₂ content. In general, the difference in the design of the plasma torch and its power may affect the residence time splitting of the CO₂, and thus the CO concentration may differ. Also, different feedstocks may influence the CO concentration due to their different compositions and process conditions, i.e., the dominance of the Boudouard (R2) and partial oxidation reactions. In some cases, the content of other syngas compounds in the present study, such as H₂, is similar to (RDF [18] and electrical switch waste [26]) or lower than (RDF [25] and keyboard plastic waste [37]) the results reported in the literature. This was mainly attributed to the different process conditions, such as the extra gasifying agents (H₂O, O₂, etc.) used and the feedstock elemental composition, affecting the dominant chemical reactions.

The change in the CO₂-to-C ratio had no significant effect on the lower heating value of the syngas (LHV_{syngas}). As the CO₂-to-C ratio increased from 1.26 to 2.34, the LHV_{syngas} remained stable in the range of 13.5–13.88 MJ/Nm³, as shown in Figure 4. This was also the case for the H₂/CO ratio, which was in the range of 0.38–0.44. The LHV_{syngas} obtained in this research was higher compared to the CO₂ plasma gasification of RDF and bakelite-based plastic waste and equal to the pure CO₂ plasma gasification of RDF [18,26,38]. The pure CO₂ plasma gasification of the computer keyboard plastic waste in the DC arc plasma of 2 kW power yielded a higher syngas calorific value of 16.46 MJ/Nm³ [37]. However, the authors reported that the operating conditions were not optimum for this process case. At the best process performance (P = 1.12 kW, 0.5 lpm of CO₂, and 40 g/10 min of feedstock mass flow rate), the LHV_{syngas} was 15.8 MJ/Nm³.

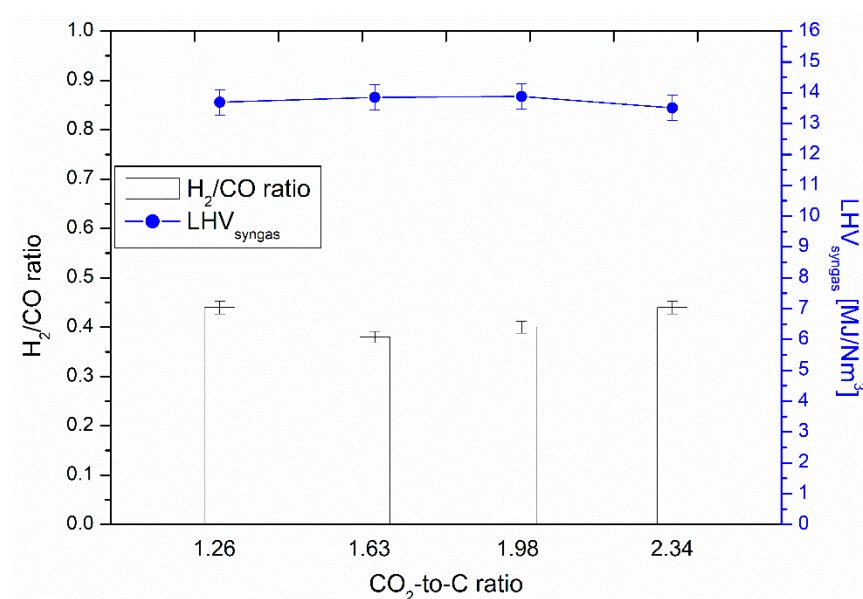


Figure 4. Effect of the CO₂-to-C ratio on the LHV of syngas and the H₂/CO ratio.

Additionally, the tar content obtained in the syngas was relatively high and ranged from 23 to 32.9 g/Nm³. The higher the CO₂-to-C ratio, the lower the tar concentration determined. This could be explained by the higher plasma torch power transferred to the reactor while converting the feedstock, which increased by around 40%, i.e., from 40.6 to 68.4 kW. The increased plasma torch power was mainly affected by the increased mass flow rate of the CO₂ used as a plasma-forming gas. Thus, with the electric arc current kept constant, the increased plasma-forming gas flow rate affected the increase of the arc's voltage drop. As a result, the plasma torch power increased. The most abundant tar compounds were benzene (12.6–16.75 g/Nm³), toluene (1.69–5.0 g/Nm³), and naphthalene (3.0–3.75 g/Nm³), altogether comprising more than 75–77% of the total tar composition. However, the obtained tar content was high and would require further syngas processing and upgrading. The reason for the higher tar concentration was that the generated plasma itself was the only heat source heating the whole reactor. No other additional electric heating of the reactor walls was used in the process, which would have increased the energy penalty and ultimately lowered the CGE. Thus, the design of the plasma gasifier could affect the final desirable gaseous product but could be improved by installing a low-power additional plasma torch (up to a few kilowatts) or catalytic cleaning. In general, plasma gasification is a method for high-purity syngas production with extremely low tar content [39]; however, it requires balancing the process performance and the energy efficiency. For instance, Hlina et al. reported a tar content of 10 mg/Nm³ after wood biomass argon/water vapor plasma gasification. Agon et al. [40] used the same plasma gasifier as the previous researcher cited, but this time for RDF plasma gasification using different gasifying agents (CO₂ + O₂, H₂O, CO₂ + H₂O, and O₂ + H₂O). In this case, the total tar content in the syngas was in the range of 0.132–0.543 g/Nm³, with the most abundant tar compounds being naphthalene, acenaphthylene, and phenanthrene [18]. The plasma gasifier was electrically preheated for 24 h to 1200 K to prevent the destruction of the ceramic coating in all the reported cases. In this research, the reactor was not externally preheated; all of the heat came from the plasma torch.

The dependence of the CO₂-to-C ratio on the syngas yield, as well as the H₂ and CO yield, is shown in Figure 5. As the CO₂-to-C ratio increased from 1.26 to 2.34, the syngas yield produced from 1 kg of FFP2 plastic feedstock increased from 1.77 Nm³ to 3.13 Nm³. This was mainly attributed to the increased mass flow rate of the CO₂ used as the plasma gas and the gasifying medium, thus increasing the CO content and yield in the producer gas due to the CO₂-splitting (R1) and Boudouard (R2) reactions. The H₂ yield constantly increased, possibly due to the water–gas shift (R3) and thermal feedstock cracking reactions. The obtained syngas yield corresponds well with the CO₂ plasma gasification of RDF, as reported by [38]. The authors claim the highest producer gas yield of 3.44 Nm³/kg_{RDF} at the RDF feeding rate of 11.3 kg/h. The higher syngas yields could be a reason for the higher carbon-to-gas conversion rate, which the authors mentioned above also reported. The highest yields of the separate syngas compounds, H₂ and CO, were obtained at a CO₂-to-C ratio of 2.34, reaching their highest values of 43.24% and 68.59%, respectively. Agon et al. [18] reported a similar CO yield of 67% using a CO₂ + O₂ mixture as a gasifying agent (cases 1, 2). However, the reported H₂ yield of 76% was around 43% higher than in the present study. This could be due to the DC water/argon plasma torch used in their research, where water was the main plasma-forming gas, but a CO₂ + O₂ gas mixture was added as the main gasifying agent. Thus, some of the H₂ could have come from the water-splitting reaction by the electric arc.

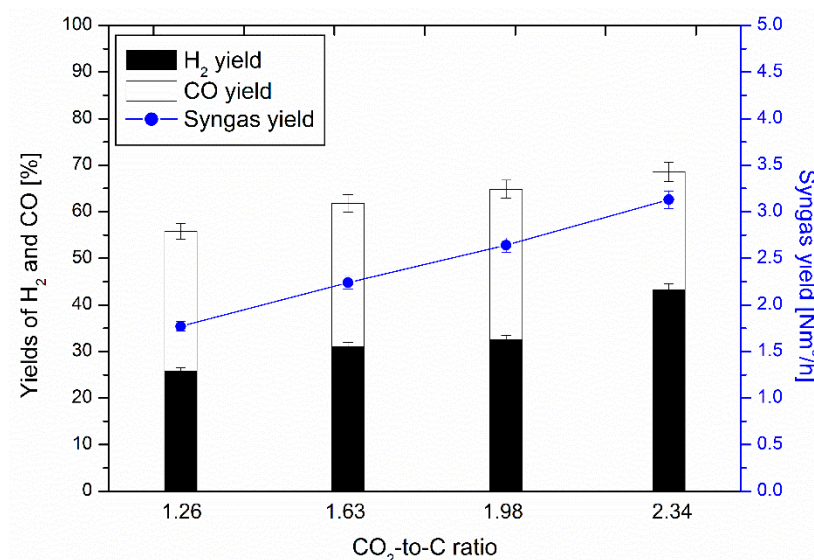


Figure 5. Effect of the CO₂-to-C ratio on the yield of syngas, H₂, and CO.

The effect of the CO₂-to-C ratio on the carbon conversion efficiency (CCE) and cold gas efficiency (CGE) is shown in Figure 6. As the CO₂-to-C ratio increased from 1.26 to 2.34, the CCE and the CGE increased from 41.1% and 30.57% to 70.6% and 47.8%, respectively. The CCE increase was mainly affected by the increase in the plasma torch power from 40.6 kW to 68.4 kW, meaning that more energy was transferred directly from the plasma stream to the FFP2 feedstock to be gasified. The increased mass flow rate of the CO₂ used as the plasma gas and gasifying medium affected the higher plasma torch power. This influenced the increase in the voltage drop of the plasma torch electric circuit without the need to increase the current, which would have led to a decrease in the coefficient of thermal efficiency. As a result, the plasma stream transferred more power at a higher efficiency to the reactor with lower losses to the plasma torch cooling water. Higher CCE values were reported by Agon et al. [18] and Sikarwar et al. [38] gasifying RDF: 83% and 100%, respectively. It could be that the plasma torch power was much higher in the referenced cases, exceeding 90–120 kW, and that the reactor was preheated before the experiments, and thus more energy was transferred to the reactor, and/or there were some methodological differences. In our case, for precision, not only was the amount of carbon present in the FFP2 feedstock and from the CO₂ as the gasifying agent taken into consideration, but the amount of carbon that remained in the ash/char and tar was also evaluated.

By comparing the CGEs, it can be determined that, in our case, the value of 47.8% is higher than that reported in other referenced plasma gasification cases using CO₂ or its mixtures as gasifying agents for RDF and plastic waste conversion [18,25,26,37,38]. However, the difference is insignificant, reaching CGE values of up to 46%. In one case, Li et al. [26] reported that the highest obtained CGE value was 49.9% during CO₂ plasma gasification of RDF. Despite this, the process conditions were not optimal. At the optimal conditions, the CGE was only 39.6%. Generally, CGE in lab-scale plasma gasification is not an optimized performance parameter compared to the industrial process, where the process heat would have been recovered and reused. However, the CGE of plasma gasification is still lower than that of conventional gasification, where the CGE is usually in the range of 60–80%. This is because plasma needs electricity to run the allothermal gasification process. Thus, to increase the CGE in plasma gasification, wastes/feedstocks with a higher calorific value should be used. Also, the process should be performed at the lowest possible plasma torch power so as not to ruin the CCE. Even the CGE of plasma gasification is lower compared to conventional gasification, but the plasma method's versatility compensates for this in

the ability to convert various heterogeneous feedstocks and to easily control and adjust the process parameters in situ, such as the feedstock feeding rate, plasma power, gasifying agent, etc.

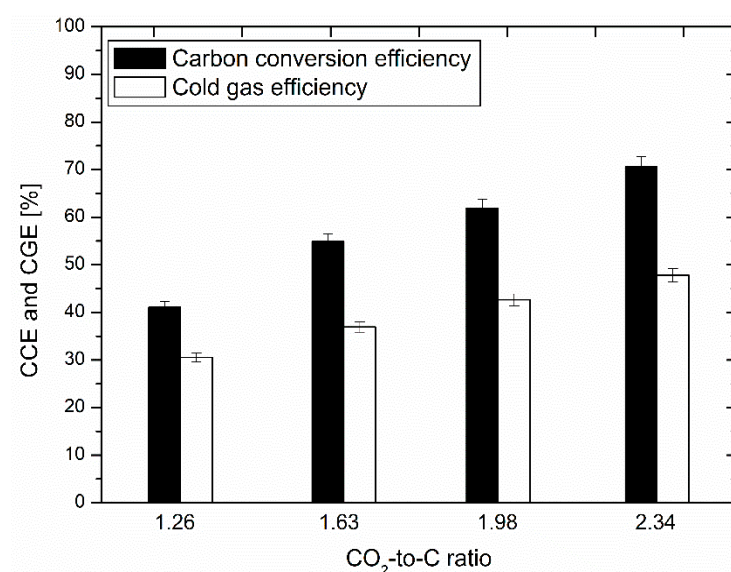


Figure 6. Effect of the CO₂-to-C ratio on the CCE and CGE.

3.3. Plasma Gasification Process Mass and Energy Balance

The literature lacks detailed data on plasma gasification process mass and energy balance. Therefore, performing a more thorough analysis and comparison is complicated. For instance, Striugas et al. [41] reported a detailed gasification process mass and energy balance in which the thermal plasma was coupled with a downdraft gasifier, converting sewage sludge to syngas. However, the plasma was used to upgrade the syngas. An integrated medical waste-to-energy concept was numerically investigated, coupling plasma gasification for medical waste conversion with MSW incineration; multiple approaches were investigated, including energy analysis, exergy analysis, and economic analysis [42]. The same authors also reported a numerical investigation of poly-generation systems combining sewage sludge pyrolysis and medical waste plasma gasification, providing an energy flow diagram with exergy and economic analysis [22]. Despite this, the performance of the plasma-alone gasification process requires a deeper investigation, particularly the mass and energy balance. Thus, this subsection describes in detail the plasma gasification mass and energy balance of gasifying medical plastic waste to overcome the lack of data availability.

Figure 7 shows the Sankey diagram of the mass balance of the FFP2 pellets' CO₂ plasma gasification to syngas at the highest CO₂-to-C ratio of 2.34. At this ratio, the highest plasma gasification process efficiency was achieved. The ultimate desired product, syngas, comprised more than 92.0% of the final mass composition. Almost 8.0% of the final product composition was associated with losses forming by-products. The main by-product constituent was solid char/ash, which comprised around 74.5% of the total by-product mass, followed by tars (17.1%) and condensate (8.3%). In general, the plasma reduced solid residues to less than 18 wt% of the primary input FFP2 feedstock material. As described above, the tar content could be reduced by installing an extra-small-power plasma torch to thermally crack the heavy hydrocarbons into lighter ones. Other methods, such as physical treatment or catalytic reforming, could also be considered [43].

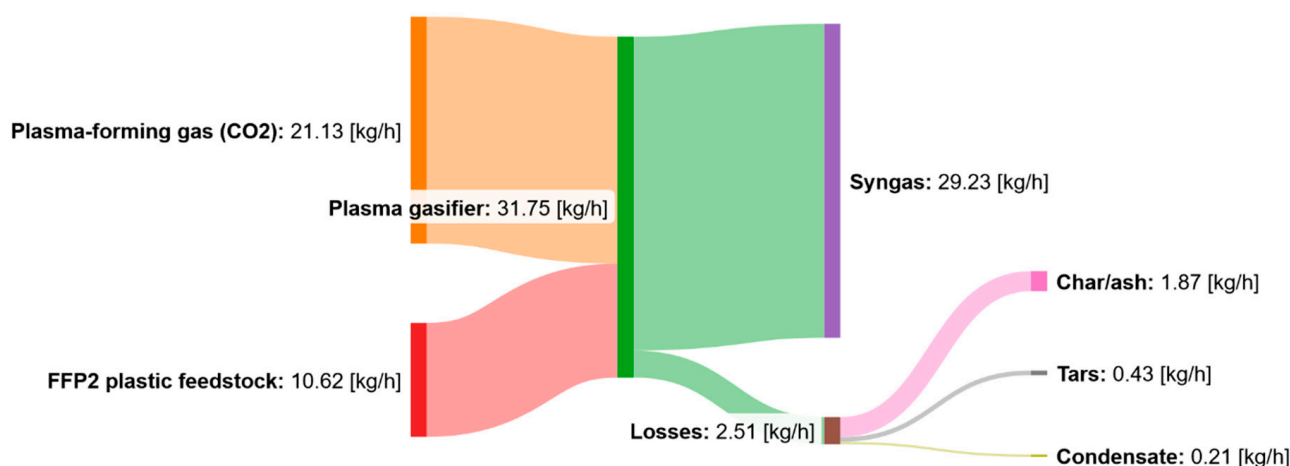


Figure 7. Mass balance of the FFP2 plastic pellets gasification in the CO₂ thermal plasma at a CO₂-to-C ratio of 2.34.

The Sankey energy flow diagram of gasifying the FFP2 plastic pellets in the CO₂ thermal plasma environment was defined at the same CO₂-to-C ratio of 2.34 as for the mass balance case and is shown in Figure 8. The highest energy content of 69.15% after the plasma gasification was obtained in the syngas, followed by the energy loss due to radiation and reactor cooling by air (21.3%), and further energy was stored in the ash/char (7.2%), tars (2.2%), and condensate (0.047%). The option of further syngas utilization after the conditioning step is theoretically proposed for use in the gas microturbine and/or internal combustion engine (ICE). As the syngas-to-electricity conversion efficiency in the microturbine is higher due to the microturbine's higher conversion performance efficiency (20% compared to the ICE efficiency of 17%), the latter case option is shown in Figure 8 [44]. Around 36% (or 24.48 kWh) of the electrical energy could be recovered to power the plasma torch (this was lower for the ICE case, at 30.4% or 20.8 kWh). The remaining energy could be recovered as heat stored in hot water. The overall process efficiency of converting the syngas to electricity via the microturbine was 12.7%, and was slightly lower for the ICE at 10.8%. For comparison, Karellas et al. [45] reported a 12–18% efficiency in electricity production by coupling biomass gasification with a Capstone C30 turbine with an electrical efficiency of 26% and a power output of 30 kW. Bartela et al. [46] integrated a 75 kW_{el} Stirling engine (engine efficiency 26.8%) into biomass gasification for combined heat and power (CHP) production and achieved 11.7% process efficiency. Tamošiūnas et al. [29] performed biomass (wood pellets) gasification to syngas in thermal water vapor plasma. An 11.5% electrical conversion efficiency using microturbine and 9.77% using ICE were potentially obtained.

The optional method of converting syngas to fuels via the Fischer–Tropsch process was not considered, as the H₂/CO ratio would require a significant syngas upgrade via the additional water–gas shift process to reach the optimal ratio 2. This method would be more beneficial if water vapor were used as a gasifying agent, yielding a higher H₂ concentration in the syngas.

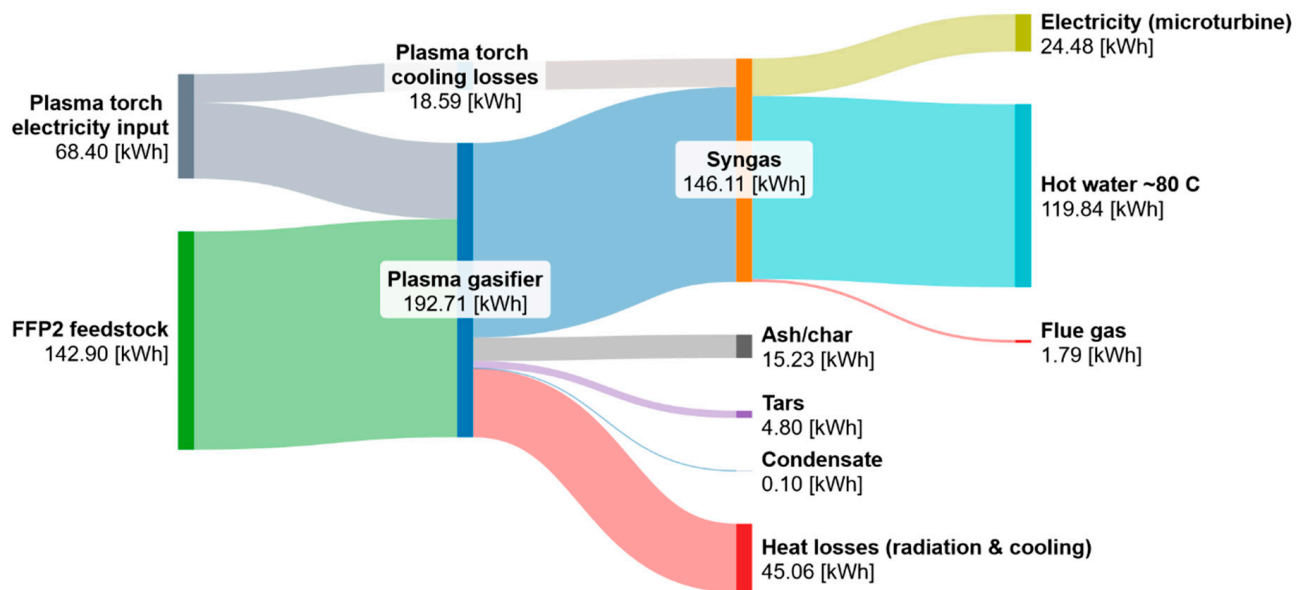


Figure 8. Sankey diagram for the energy flow of the FFP2 plastic pellets gasification in the CO₂ thermal plasma at a CO₂-to-C ratio of 2.34.

4. Conclusions

This experimental work investigated the plasma gasification of FFP2 respirator face masks to syngas in a greenhouse gas (CO₂) environment to reduce the environmental pollution caused by plastic waste and anthropogenic CO₂ emissions and simultaneously to recover energy. A DC thermal arc plasma torch operating at atmospheric pressure generated a plasma stream for allothermal gasification. The effect of the CO₂-to-C ratio on the gasification performance efficiency was analyzed. The highest CO₂ plasma gasification efficiency was obtained at the highest CO₂-to-C ratio of 2.34. At these conditions, the main syngas components (H₂ + CO) comprised 80.46 vol.% (H₂: 24.62 vol.% and CO: 55.84 vol.%), the LHV_{syngas} was 13.88 MJ/Nm³, the syngas yield was 3.13 Nm³/kg_{FFP2}, the tar content in the syngas was 23 g/Nm³, the CCE was 70.6%, and the CGE was 47.8%. Additionally, the plasma gasification process mass and energy balance were analyzed.

The results show that CO₂ plasma gasification is a promising thermochemical treatment method for more sustainable plastic waste conversion and the utilization of greenhouse gases, such as carbon dioxide. Future work will investigate CO₂ splitting to CO using this system and maximize the gasification performance to determine the optimal process parameters.

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Abbreviations

The following abbreviations are used in this manuscript:

DC	direct current
GHG	greenhouse gas
CCE	carbon conversion efficiency
CGE	cold gas efficiency
LHVsyngas	lower heating value of syngas
WHO	World Health Organization
MSW	municipal solid waste
RDF	refuse-derived fuels
CCU	carbon capture and utilization
PP	polypropylene
PE	polyethylene
SER	specific energy requirement

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